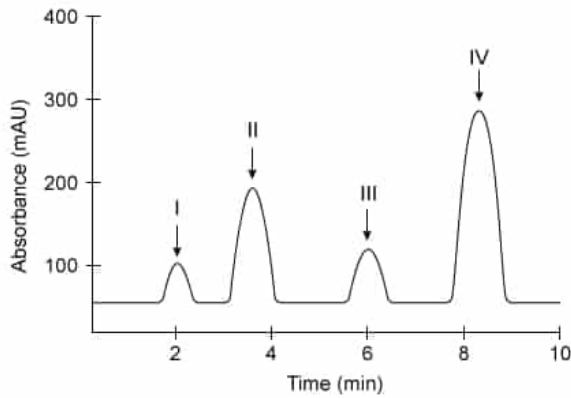




The above chromatogram shows the separation of a mixture containing a methyl ester, a carboxylic acid, an alcohol, and an alkane. Each parent chain has the same number of carbons. If normal-phase, high-performance liquid chromatography is used to separate the mixture components, which peak corresponds to the alkane?

- ✓ ☒ A. I [76%]
☐ B. II [4%]
☐ C. III [3%]
☐ D. IV [15%]

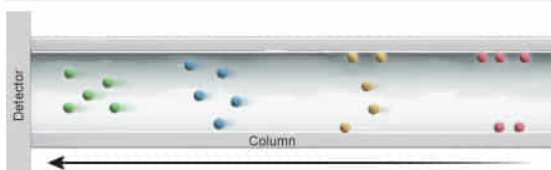
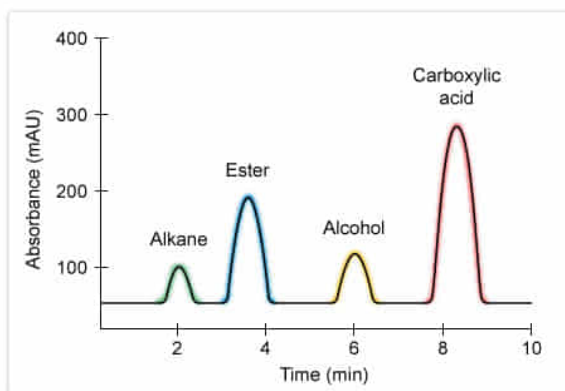
Correct

76% answered correctly

Explanation:

Normal-phase HPLC

Polar stationary phase (silica) and nonpolar mobile phase



Nonpolar

- Interacts more with mobile phase
- Shorter retention time

Polar

- Interacts more with stationary phase
- Longer retention time

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High-performance liquid chromatography (HPLC), a purification technique ideal for

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small sample sizes, separates compounds based on **polarity**. The instrumentation consists of a sample injector, column (stationary phase), solvent under pressure (mobile phase), detector, and computer for data acquisition.

One of **two types of columns**—**normal-phase (NP)** or **reverse-phase (RP)**—may be used. **NP-HPLC** consists of a **polar stationary phase** (typically silica) and a comparatively **nonpolar mobile phase**. In contrast, RP-HPLC has a nonpolar stationary phase (eg, C18 alkyl hydrocarbon) and a comparatively polar mobile phase.

If a mixture is separated by NP-HPLC, the **less polar compounds elute before more polar compounds** because the more polar compounds have a stronger **interaction** with the polar stationary phase, causing them to have a longer retention time. A mixture containing an ester, a carboxylic acid, an alcohol, and an alkane separated by NP-HPLC elutes in the following order (nonpolar to polar): alkane, ester, alcohol, and carboxylic acid. Therefore, **peak I** (shortest retention time) corresponds to the **alkane**.

(Choices B, C, and D) The ester, alcohol, and carboxylic acid (peaks II, III, and IV, respectively) are all *more polar* than the alkane. These molecules interact more readily with the polar stationary phase than with the nonpolar mobile phase; therefore, all three have longer retention times than the alkane (peak I).

Educational objective:

In high-performance liquid chromatography (HPLC), two types of columns—normal-phase (NP) and reverse-phase (RP)—can be used, depending on the polarities of the compounds being separated. NP-HPLC consists of a polar stationary phase and a nonpolar mobile phase whereas RP-HPLC consists of a nonpolar stationary phase and a polar mobile phase. Molecules with polarity similar to the stationary phase interact with it more and have longer retention times.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402752

System:
Separation Techniques, Spectroscopy, and Analytical
Methods